MOHAMMAD ROHAAN

BSCS Student (FAST) | Software Engineering, Systems & Performance | ML | AI | CNN |CV Interest (Learning Path)

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OBJECTIVE

7th-semester **Computer Science** student with strong delivery in systems **programming, performance analysis, concurrency, and data-heavy applications**. Built production-style projects across .NET/SQL, Java desktop, embedded robotics, and algorithmic systems (version control, Merkle proofs, HPC kernels). Targeting high-impact SWE roles with a growth path toward Applied ML / computer vision; actively deepening skills in modern ML workflows (coursework + hands-on projects — CNN/CV/NLP foundations in progress).

EDUCATION

FAST University, Islamabad — BSCS | Expected graduation: Jan 2027 | 7th semester
Relevant coursework: Data Structures & Algorithms, OS, Computer Networks, Embedded Systems, Numerical Computing / HPC, Database Systems, DAA, Robotics (MATLAB / kinematics).

SELECTED PROJECTS

HPCG Kernel Optimization — Improved SpMV/SymGS-related performance; integrated with reference HPCG; benchmarked, analyzed memory behavior, documented results. (Numerical / ML-relevant linear algebra.)

Git Lite (C++) — Mini version control: DAG/commits, hashing, diff/merge strong CS fundamentals.

Airline Booking Simulator (C, OS) — pthreads + sync primitives; deadlock-aware design; stress testing.

Robotic Car Synchronization — ESP32 + Python coordination; sensors, real-time control themes.

Enterprise Network Simulation (Cisco PT) — Multi-area OSPF, redistribution, DHCP/NAT/ACLs

Inventory / Stock Management (Java + SQL) — Auth, real-time stock, reporting (application delivery).

Text Similarity / Plagiarism-style project — Preprocessing + similarity scoring; classical NLP-adjacent, not deep learning.

Large-Scale Merkle Tree Verification (DAA) — Integrity proofs on 1M+ review records; tamper detection; benchmarking hash/proof latency and memory.

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, SQL, Assembly x86

Systems & performance: Threads, mutexes/semaphores, HPCG/SpMV, numerical methods

Data & backend: SQL Server, T-SQL, schema design, ETL/reporting (large datasets)

Application dev: .NET, Java Swing/JavaFX, Git, Linux, shell scripting

Embedded & robotics: Arduino, ESP32, sensors, PID/control (as per project), ROS (beginner)

Networking (lab): Cisco Packet Tracer — OSPF/EIGRP/RIP, NAT, ACLs, DHCP (course projects)

ML / AI (in progress —): Python ML fundamentals (course/self-study); interest in CNNs & CV;

TARGET POSITION

- Software Engineer / Backend / Systems: lead with .NET/SQL + OS + performance projects.
- ML Engineer (junior/intern): lead with Python + numerics . Intrested in ___ NLP | CV | CNN | ML | AI